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Question 1 of 40

Which of the following sorting algorithms in its typical implementation gives best performance when applied on an array which is sorted or almost sorted (maximum 1 or two elements are misplaced).

- ☐ A. Quick Sort
- ☐ B. Heap Sort
- ☐ C. Merge Sort
- ☒ D. Insertion sort

Questions

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
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Question 2 of 40

What is the best time complexity of bubble sort?

- ☐ A. $O(n)^2$
- ☐ B. $O(n) \log n$
- ☒ C. $O(n)$
- ☐ D. $O(n) (\log n)^2$

Questions

1	2	3	4	5	6	7	8	9	10
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Question 3 of 40

Consider a situation where swap operation is very costly. Which of the following sorting algorithms should be preferred so that the number of swap operations are minimized in general?

☐ A. Heap Sort

☒ B. Selection sort

☐ C. Insertion sort

☐ D. Merge sort

Questions

1	2	3	4	5	6	7	8	9	10
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Question 4 of 40

Which of the following algorithm design technique is used in Quick sort?

☐ A. Greedy method

☐ B. Backtracking

☐ C. Dynamic programming

☒ D. Divide and Conquer

Questions

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11	12	13	14	15	16	17	18	19	20
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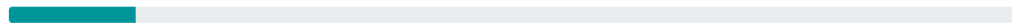
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Question 5 of 40

Selection sort algorithm design technique is an example of

☒ A. Greedy Method

☐ B. Divide and Conquer

☐ C. Dynamic programming

☐ D. Backtracking

Questions

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21	22	23	24	25	26	27	28	29	30
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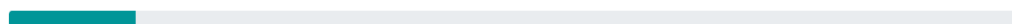
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Question 6 of 40

The result of evaluating postfix expression $539*74-/+62*-$ is

☐ A. 16

☐ B. 2

☐ C. -7

☐ D. 3

Questions

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
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Question 7 of 40

What would be the time complexity of push and pop operation if, the stack implemented using linked list?

- ☐ A. $O(1)$ for insertion $O(n)$ for deletion
- ☒ B. $O(1)$ for insertion $O(1)$ for deletion
- ☐ C. $O(n)$ for insertion $O(1)$ for deletion
- ☐ D. $O(n)$ for insertion $O(n)$ for deletion

Questions

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
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Question 8 of 40

If one dimensional array $A[\text{MAXSIZE}]$ is used to implement two stacks provided two stacks grow from opposite ends of the array. Variables top1 and top2 ($\text{top1} < \text{top2}$) point to the location of the topmost element in each of the stacks, then which of the followings statements must be true?

- I. If the space is to be used efficiently, the condition for stack full is $\text{top1} = \text{top2} - 1$
- II. The condition for stack full is $\text{top1} + \text{top2} = \text{MAXSIZE}$

- ☐ A. Only I is true
- ☐ B. Only II is true
- ☐ C. Both I and II are true
- ☐ D. Both I and II are false

Questions

1	2	3	4	5	6	7	8	9	10
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21	22	23	24	25	26	27	28	29	30
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Question 9 of 40

The _____ data structure is used to check whether an arithmetic expression has balanced parenthesis or not

☐ A. Queue

☐ B. List

☐ C. Tree

☒ D. Stack

Questions

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Question 10 of 40

Which of the following is the disadvantage of the array?

☐ A. If user wishes to store multiple values of similar type then the Array can be used and utilized efficiently.

☐ B. In arrays, the elements can be accessed randomly by using the index number.

☐ C. Insertion and deletion are quite difficult in an array as the elements are stored in consecutive memory locations and the shifting operation is costly.

☐ D. Arrays allocate memory in contiguous memory locations for all its elements.

Questions

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Question 11 of 40

Which one of the following is an application of Queue Data Structure?

- ☐ A. Spooling in printers
- ☐ B. Queues in routers/ switches
- ☐ C. Evaluation of Arithmetic Expressions
- ☐ D. All of the above
- ☒ E. Both A and B

Questions

1	2	3	4	5	6	7	8	9	10
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Question 12 of 40

Which of the following is true about linked list implementation of queue?

- ☐ A. In push operation, if new nodes are inserted at the beginning of linked list, then in pop operation, nodes must be removed from end.
- ☐ B. In push operation, if new nodes are inserted at the end of linked list, then in pop operation, nodes must be removed from beginning.
- ☐ C. Queue data structure can be used to implement least recently used (LRU) page fault algorithm and Quick sort algorithm.
- ☒ D. All of the above

Questions

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
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Question 13 of 40

Consider the following pseudo code. Assume that IntQueue is an integer queue. What does the function fun do?

```
void fun(int n)
{
    IntQueue q = new IntQueue();
    q.enqueue(0);
    q.enqueue(1);
    for (int i = 0; i < n; i++)
    {
        int a = q.dequeue();
        int b = q.dequeue();
        q.enqueue(b);
        q.enqueue(a + b);
        print(a);
    }
}
```

- ☐ A. Prints numbers from 0 to n-1
- ☐ B. Prints numbers from n-1 to 0
- ☒ C. Prints first n Fibonacci numbers
- ☐ D. Prints first n Fibonacci numbers in reverse order.

Questions

1	2	3	4	5	6	7	8	9	10
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Question 14 of 40

.....is an abstract data type that generalizes a queue, for which elements can be added to or removed from either the front or rear end

- ☐ A. Double ended queue
- ☐ B. Deck
- ☒ C. Deque
- ☐ D. only A is correct
- ☐ E. Both A and C are correct

Questions

1	2	3	4	5	6	7	8	9	10
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Question 15 of 40

What will be queue full condition for circular queue if implemented using static array, provided rear and front are initially -1?

☐ A. rear==SIZE

☒ B. rear==SIZE-1

☐ C. (rear+1)%size==front

☐ D. None of the above

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Questions

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Question 16 of 40

Which of the following statement is true about the complete binary tree?

I. In complete binary tree, every level must be filled from left to right direction

II. Every full binary tree can be considered as a complete binary tree but reverse is not true

☐ A. Only I is true

☒ B. Only II is true

☐ C. Both I and II are True

☐ D. None of the above

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Questions

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Question 17 of 40

Which one of the following is an application of Graph Data Structure?

☐ A. Electronic circuit simulation

☐ B. Path Optimization Algorithms.

☐ C. Scientific Computations

☒ D. All of the above

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Questions

1	2	3	4	5	6	7	8	9	10
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Question 18 of 40

In binary tree if non-leaf node has non-empty left sub-tree and non-empty right sub-tree then such binary tree is called as

☒ A. Strictly binary tree

☐ B. Complete binary tree

☐ C. Full binary tree

☐ D. Skewed binary tree

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Questions

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Question 19 of 40

Suppose the numbers 80 110 120 60 40 70 90 100 85 are inserted in that order into an initially empty binary search tree. The binary search tree uses the usual ordering on natural numbers. What is the in-order traversal sequence of the resultant tree?

☐ A. 80 60 40 70 110 90 85 100 120

☐ B. 40 60 70 80 85 90 100 110 120

☐ C. 40 70 60 85 100 90 120 110 80

☐ D. 40 60 70 85 90 100 110 120 80



Finish

Questions

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Question 20 of 40

Given an undirected graph G with 5 vertices and 7 edges, the sum of the degrees of all vertices is:

☐ A. 7

☐ B. 5

☐ C. 10

☐ D. 14



Finish

Questions

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